

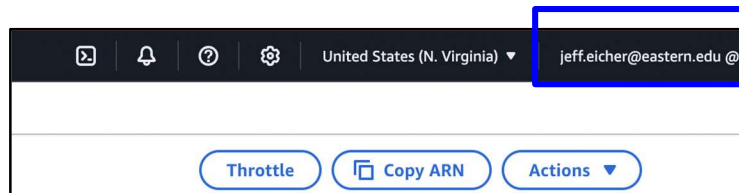


DTSC 671: AI Solutions in the Cloud

SUBMISSION TEMPLATE: Sentiment Analysis with Amazon Comprehend

Select *File > Make a copy* to edit the template and add your screenshots. Download the completed template as a PDF document to submit to Brightspace.

In every screenshot, include your AWS username that appears in the upper right corner. Example:

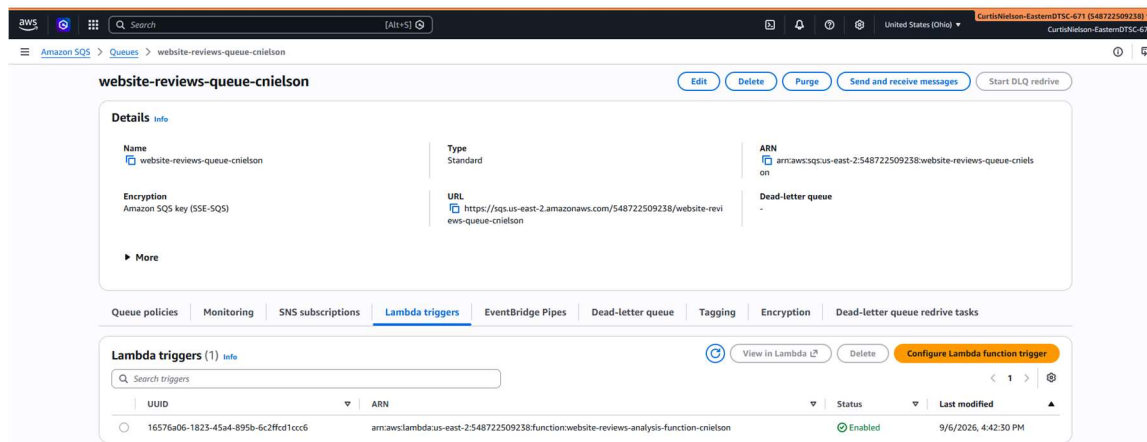


If any one screenshot is missing this identification, the entire assignment will be scored no higher than 70%.

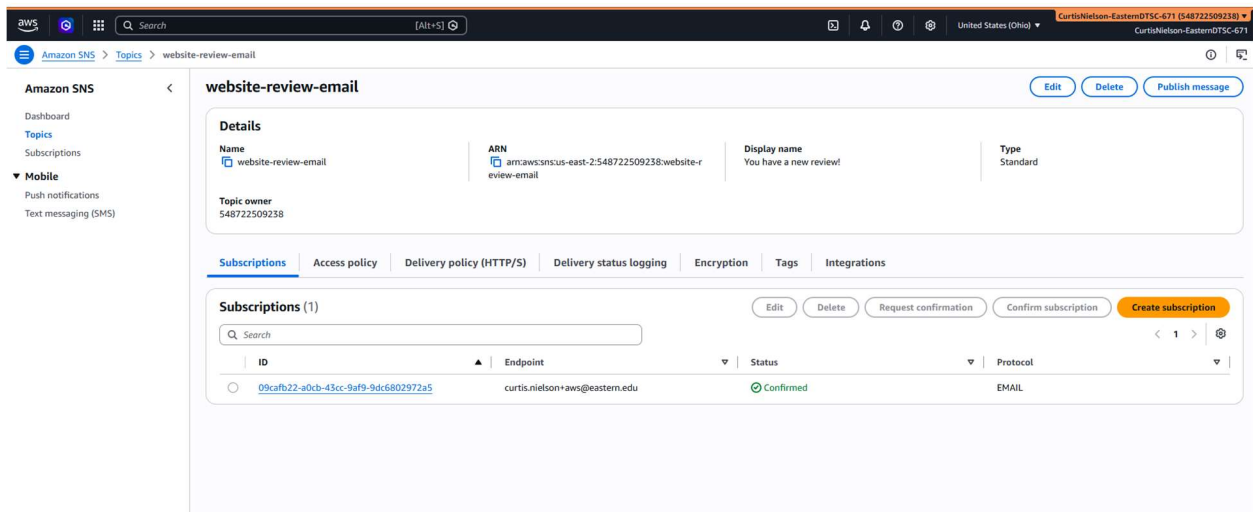
Points totals are provided at the start of each question. You score either full credit or 0 points; there is no partial credit for each question.

Curtis Nielson
DTSC-671

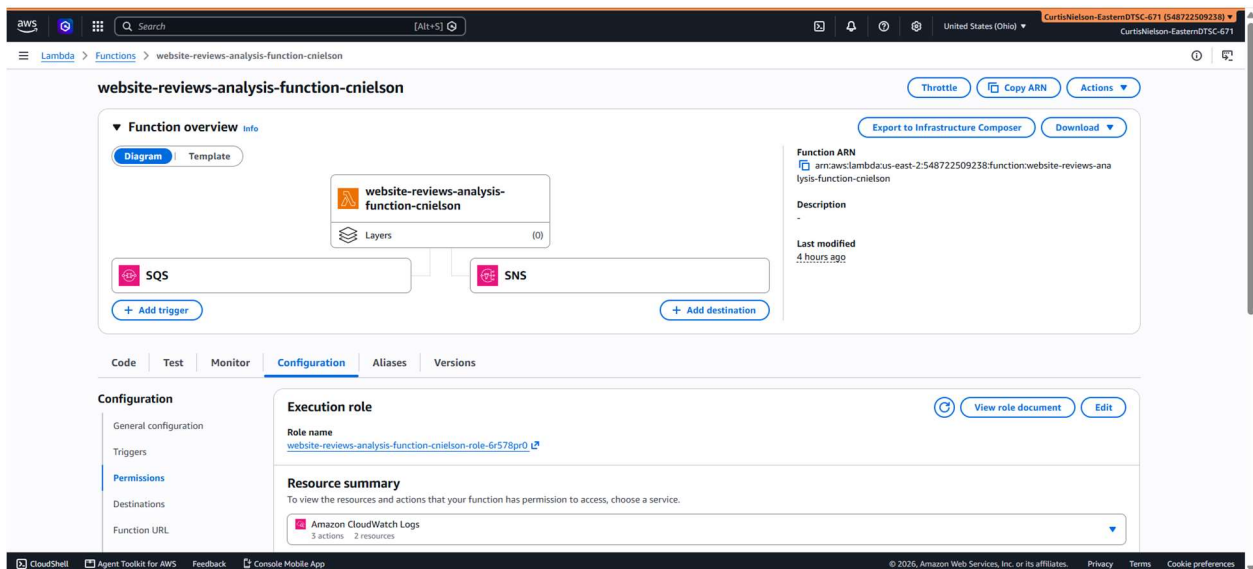
- 1) [10 points] In the space below, provide a screenshot of the SQS queue.



2) [10 points] In the space below, provide a screenshot of the SNS topic.

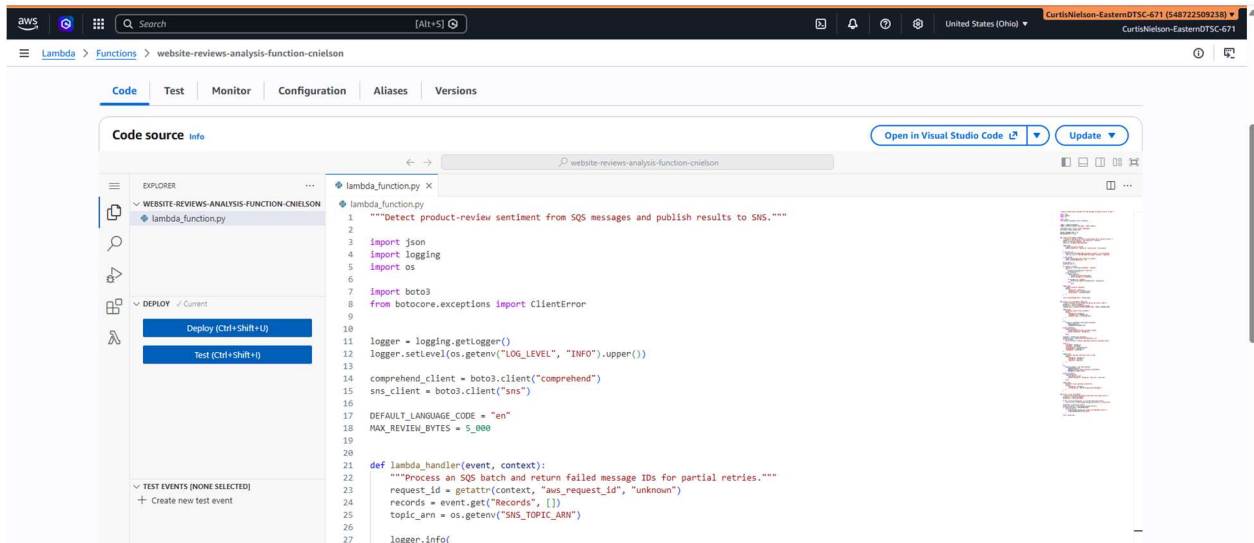


3) [10 points] In the space below, provide a screenshot of the Lambda diagram, which shows the SQS trigger and the SNS destination.



4) [10 points] In the space below, provide both a screenshot of the beginning of your Lambda code with your account showing, then copy-paste your full lambda code below the screenshot.

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```
"""Detect product-review sentiment from SQS messages and publish results to SNS."""
```

```
import json
import logging
import os
```

```
import boto3
from botocore.exceptions import ClientError
```

```
logger = logging.getLogger()
logger.setLevel(os.getenv("LOG_LEVEL", "INFO").upper())
```

```
comprehend_client = boto3.client("comprehend")
sns_client = boto3.client("sns")
```

```
DEFAULT_LANGUAGE_CODE = "en"
MAX_REVIEW_BYTES = 5_000
```

```
def lambda_handler(event, context):
    """Process an SQS batch and return failed message IDs for partial retries."""
    request_id = getattr(context, "aws_request_id", "unknown")
    records = event.get("Records", [])
    topic_arn = os.getenv("SNS_TOPIC_ARN")

    logger.info(
        "Lambda invocation started",
        extra={"request_id": request_id, "record_count": len(records)},
    )
```

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```

if not topic_arn:
    logger.error("SNS_TOPIC_ARN environment variable is not configured")
    raise RuntimeError("SNS_TOPIC_ARN environment variable is required")

if not records:
    logger.warning("SQS event contains no records")
    return {"batchItemFailures": []}

failed_items = []
processed_count = 0

for record in records:
    message_id = record.get("messageId", "unknown")
    try:
        process_sqs_record(record, topic_arn)
        processed_count += 1
    except Exception:
        logger.exception(
            "Failed to process SQS message",
            extra={"message_id": message_id},
        )
        if message_id != "unknown":
            failed_items.append({"itemIdentifier": message_id})
        else:
            raise

logger.info(
    "Lambda invocation completed",
    extra={
        "request_id": request_id,
        "processed_count": processed_count,
        "failed_count": len(failed_items),
    },
)

return {"batchItemFailures": failed_items}

def process_sqs_record(record, topic_arn):
    """Detect sentiment for one review and publish the result to SNS."""
    message_id = record["messageId"]
    review_text = extract_review_text(record)
    language_code = os.getenv("REVIEW_LANGUAGE_CODE",
    DEFAULT_LANGUAGE_CODE)

    logger.info(
        "Analyzing product review sentiment",
        extra={
            "message_id": message_id,
            "language_code": language_code,

```

```

        "character_count": len(review_text),
    },
)

try:
    response = comprehend_client.detect_sentiment(
        Text=review_text,
        LanguageCode=language_code,
    )
except ClientError:
    logger.exception(
        "Amazon Comprehend detect_sentiment failed",
        extra={"message_id": message_id},
    )
    raise

sentiment = response.get("Sentiment")
sentiment_score = response.get("SentimentScore", {})
if not sentiment:
    raise RuntimeError("Amazon Comprehend returned no sentiment value")

result = {
    "messageId": message_id,
    "sentiment": sentiment,
    "sentimentScore": sentiment_score,
    "languageCode": language_code,
    "reviewText": review_text,
}

logger.info(
    "Sentiment detected; publishing result to SNS",
    extra={
        "message_id": message_id,
        "sentiment": sentiment,
        "topic_arn": topic_arn,
    },
)

try:
    publish_response = sns_client.publish(
        TopicArn=topic_arn,
        Subject=f"Product review sentiment: {sentiment}",
        Message=json.dumps(result),
    )
except ClientError:
    logger.exception(
        "SNS publish failed",
        extra={"message_id": message_id, "topic_arn": topic_arn},
    )
    raise

```

```

logger.info(
    "Sentiment result published successfully",
    extra={
        "message_id": message_id,
        "sns_message_id": publish_response.get("MessageId"),
    },
)

```

```

def extract_review_text(record):
    """Extract and validate the product-review text from an SQS record."""
    message_id = record["messageId"]
    review_text = record.get("body")

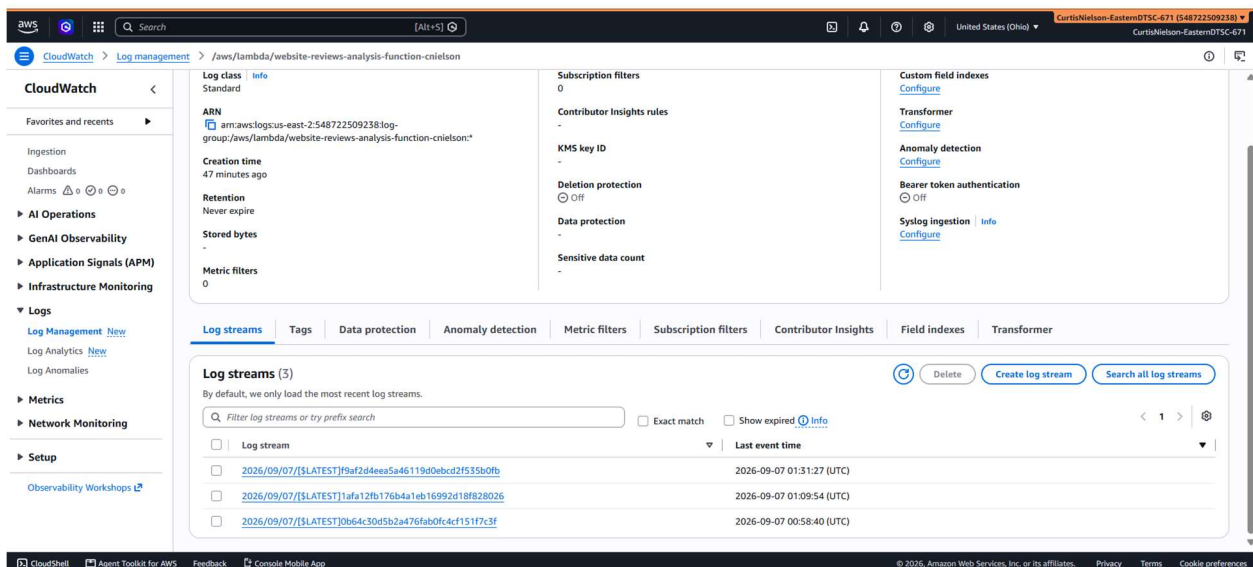
    if not isinstance(review_text, str) or not review_text.strip():
        raise ValueError(f"SQS message {message_id} contains no review text")

    review_text = review_text.strip()
    review_size_bytes = len(review_text.encode("utf-8"))
    if review_size_bytes > MAX_REVIEW_BYTES:
        raise ValueError(
            f"SQS message {message_id} exceeds the Comprehend limit of "
            f"{MAX_REVIEW_BYTES} UTF-8 bytes"
        )

    return review_text

```

- 5) [10 points] In the space below, provide a screenshot of the CloudWatch log streams for your Lambda function. Fit as many as you can in the screenshot.



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6) [10 points] In the space below, provide a screenshot of the CloudWatch stream that functioned without errors for the Lambda function.

Two successful executions are recorded below.

The screenshot displays the AWS CloudWatch Log Events console for the Lambda function `/aws/lambda/website-reviews-analysis-function-cnielson`. The interface shows a list of log events with columns for **Timestamp** and **Message**. Two successful executions are highlighted:

- Execution 1:** Occurred at 2026-09-07T00:58:33.587Z. The message indicates a successful invocation with a duration of 523.04 ms and memory usage of 128 MB. The log shows the start of the invocation, the analysis of product review sentiment, and the successful publishing of the result to SNS.
- Execution 2:** Occurred at 2026-09-07T00:58:33.588Z. The message indicates a successful invocation with a duration of 424.17 ms and memory usage of 128 MB. The log shows the start of the invocation, the analysis of product review sentiment, and the successful publishing of the result to SNS.

The console also includes a search bar, filter options, and a 'Start tailing' button. The bottom of the screen shows the AWS footer with copyright information and links to privacy and terms.

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- 7) [10 points] In the space below, provide a screenshot of the review you are testing in the SQS queue.

These three messages were tested

CeraVe hand lotion is great for treating rough, dry skin. It lasts a long time and only takes a few minutes to be absorbed into the skin. I will buy it again.

Send and receive messages

Use this page to send, retrieve and view messages, helping you experiment with various queue features.

Send message [Info](#)

✔ Your message has been sent and is ready to be received. [View sent message details](#) ✕

Message body
Enter the message to send to the queue.

CeraVe hand lotion is great for treating rough, dry skin. It lasts a long time and only takes a few minutes to be absorbed into the skin. I will buy it again.

Message group ID - optional, new [Info](#)
A group identifier for the message to allow fair processing across message groups in a standard queue.

Message group ID must be 1 to 128 characters. Valid characters are a-z, A-Z, 0-9, and punctuation (!@#\$%^&*~.-/;<+?@[\]";'{}|_`~). ([]-).

Delivery delay [Info](#)
The duration (in seconds) that SQS will postpone the initial delivery of the message. During this delay period, the message is not visible to consumers, allowing you to create a wait time before the message becomes available for processing.

0 [Seconds](#) ▼
Should be between 0 seconds and 15 minutes.

► **Message attributes - optional** [Info](#)

[Clear content](#) [Send message](#)

CeraVe hand lotion is only so so for my dry skin. It doesn't lasts long enough and feels greasy for a while until it is absorbed into the skin. It's mediocre value for the money.

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0 [Seconds](#) ▼
Should be between 0 seconds and 15 minutes.

► **Message attributes - optional** [Info](#)

[Clear content](#) [Send message](#)

CeraVe hand lotion is not very good for my dry skin. It lasts only a few hours, feels very greasy and takes a long time to be absorbed into the skin. I'm going to try something else.

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Message group ID must be 1 to 128 characters. Valid characters are a-z, A-Z, 0-9, and punctuation (["#\$%&'()*+,-./:;<=>?@[]^_`{|}~]).

Delivery delay [Info](#)
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0 [Seconds](#)

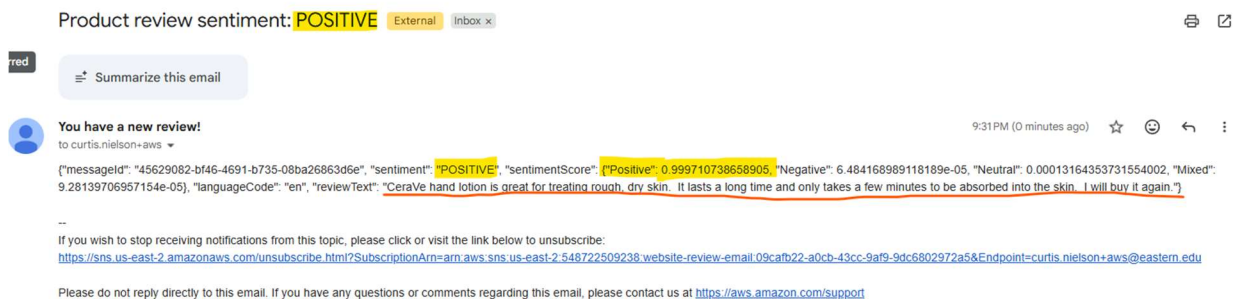
Should be between 0 seconds and 15 minutes.

► **Message attributes - optional** [Info](#)

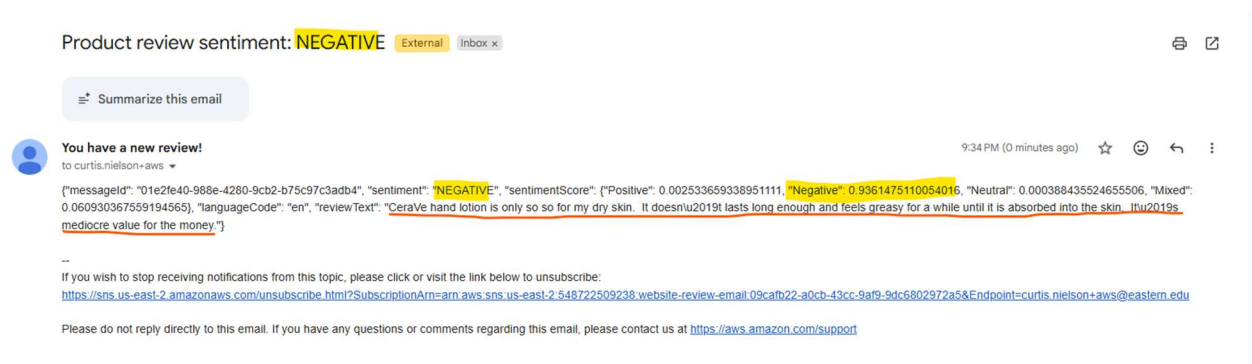
[Clear content](#) [Send message](#)

8) [30 points] In the space below, provide a screenshot of the email you receive showing both the text of the review and the sentiment (positive, negative, neutral).

The first one is clearly positive



The review where I was going for neutral came out as negative



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The final review was negative as expected

Product review sentiment: **NEGATIVE** External Inbox x

Summarize this email

You have a new review! 9:37 PM (0 minutes ago) ☆ 😊 ↶ ⋮

to curtis.nielson+aws ▾

[{"messageId": "1e18a0c9-aaae-4c33-8a39-69f96935fc83", "sentiment": "NEGATIVE", "sentimentScore": {"Positive": 0.0009395902161486447, "Negative": 0.9931668639183044, "Neutral": 0.0008953401702456176, "Mixed": 0.004998206160962582}, "languageCode": "en", "reviewText": "CeraVe hand lotion is not very good for my dry skin. It lasts only a few hours, feels very greasy and takes a long time to be absorbed into the skin. I'm going to try something else."}]

2 deleted messages in this conversation. [View messages](#) or [delete forever](#).

↶ Reply ↶ Reply all ↷ Forward 😊 🗨 Share in chat **New**

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